Multimodal Crop Health Monitoring Using Hyperspectral Imaging, IoT Sensors, and Deep Learning: A CNN-LSTM Fusion Approach

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***Abstract*—Precision agriculture has been considered to be one of the critical challenges in early detection of crop diseases, and yield prediction. In this study, an integrated multimodal framework has been proposed by combining hyperspectral imaging, IoT environmental sensors, and deep learning (DL) for comprehensive crop health monitoring. A Convolutional Neural Network (CNN) extracts spatial-spectral features from hyperspectral TIFF imagery for disease classification, while a Long Short-Term Memory (LSTM) network models temporal dependencies in IoT sensor data for predictive crop status assessment. The proposed model achieves 87.60% accuracy with a macro F1-score of 0.8752 on hyperspectral test data, while the LSTM attains 90.50% accuracy on temporal sensor sequences. The system has been compared with SVM, RGB-CNN, and 3D-CNN baselines which validate the superiority of the proposed approach. This framework addresses key gaps in precision agriculture by integrating spatial, spectral, and temporal data streams with explainable**

***Index Terms*—Precision Agriculture, Machine Learning, Sensors, Deep Learning**

I. Introduction

Global agriculture has been increased for increasing the productivity while minimizing environmental impact. Crop diseases have been accountable for 20–40% of annual yield losses worldwide, which would threaten food security and farmer livelihoods [1,2,3,4,5]. Traditional methods have been detected that would depend on agronomist visual inspection, which is subjective and labour-intensive. Precision agriculture has been introduced, where sensor based and imaging technologies are used to enable data-driven farming decisions [6,7].

Hyperspectral imaging (HSI) has been important, where the contiguous narrowband spectral data across hundreds of wavelengths would reveal biochemical and physiological plant properties with respect to RGB cameras [8]. HSI has been detected through the early-stage stress before visible symptoms appear. In addition to that, IoT deployments would generate continuous environmental data—soil moisture, temperature, humidity, and pH that influence crop health dynamics [9]. In these streams, synergistic multimodal inference have been deployed in various application domains.

A. Problem Statement

Three critical gaps have persisted in the agricultural settings. Systems have been used for imaging with respect to IoT sensors. Most deep learning approaches would treat as crop health, which would involve temporal environmental patterns. Black-box models would hinder the trust of the farmer, as explainable AI is rarely integrated into agricultural systems.

B. Contributions

The contributions of the study are:

• Develop a CNN classifier for hyperspectral crop disease detection with Grad-CAM explainability.

• Implement an LSTM network for temporal IoT sensor analysis to predict crop health status.

• Propose a multimodal late-fusion architecture integrating spatial-spectral and temporal features.

• Evaluate unimodal vs. multimodal performance using accuracy, F1-score, and AUC-ROC.

Section II reviews related work. Section III formalizes the problem. Section IV details methodology. Section V describes experimental setup. Section VI presents results. Section VII discusses findings. Section VIII concludes with future directions.

II. Literature Review

A. Hyperspectral Imaging in Agriculture

Lu et al. [10] demonstrated spectral signatures in the 400 1000 nm range, which correlates with chlorophyll content and water stress indicators. Zhang et al. [11] applied SVM on HSI for wheat disease detection by achieving 89% accuracy. Ferentinos [12] applied deep CNNs to RGB plant disease images, achieving greater than 99% accuracy on large datasets. Ahmad et al. [13] have combined HSI with 3D-CNNs for soybean stress classification, which shows the spectral spatial fusion that would outperform 2D approaches. Most of the HSI studies have ignored with concurrent environmental factors that modulate disease progression.

B. IoT and Sensor Networks in Smart Farming

Tzounis et al. [14] have been reviewed in wireless sensor networks for soil and microclimate monitoring. Kamienski et al. [15] proposed a cloud IoT platform for precision irrigation using threshold-based alerts rather than predictive modeling. Sánchez et al. [16] applied the Random Forest model on sensor data for yielding without exploiting temporal sequences.

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LSTM and recurrent architectures have not been explored for agricultural time-series forecasting.

C. Multimodal Deep Learning

Köksal et al. [17] have been used for RGB images with respect to weather data for yielding crop yield for early fusion CNNs where spatial-spectral has been exploited from temporal feature extraction. In this approach, it explicitly separates CNN (spatial-spectral) and LSTM (temporal) processing before late fusion that would allow each modality to learn optimal representations independently.

D. Explainable AI in Agriculture

Selvaraju et al. [18] introduced Grad-CAM for gradient-based visual explanations of CNN decisions. Barbedo [19] applied saliency maps for RGB plant disease images. Grad-CAM has not been widely applied to hyperspectral agricultural imagery to identify disease relevant wavelengths, which is a gap this work directly.

III. Problem Formulation

Let $I \in \mathbb{R}^{H\times W\times B}$ denote a hyperspectral image with height H, width W, and B spectral bands. Let $S = \{s_1, s_2, \ldots, s_T\}$ denote a sequence of IoT sensor readings over T time steps, where each $s_t \in \mathbb{R}^d$ contains d environmental features.

$$f(I, S) \to y \tag{1}$$

The objective is to learn a function $f(I, S)$ that predicts the crop health label

$$y \in \{\text{Healthy}, \text{Rust}, \text{Other}\} \tag{2}$$

by minimizing the multiclass cross-entropy loss.

$$f^* = \arg\min_f \ \mathbb{E}\left[L\left(f(I, S), y\right)\right] \tag{3}$$

where L denotes the categorical cross-entropy loss function, and y represents the ground-truth crop health label.

IV. Materials and Methods

A. Dataset Description

Two complementary datasets are employed. Table I summarises their key properties.

1) Hyperspectral Dataset: Hyperspectral TIFF imagery is labelled into three disease classes. Class distribution is shown to be Healthy 32% (160 images), Rust 37% (185 images), Other 31% (155 images). Stratified splitting mitigates class imbalance. All images are multi-band TIFF files processed with rasterio.

2) IoT Sensor Dataset (Simulated): : Five environmental features are generated with class, realistic uniform distributions (Table II). A sample is labelled Healthy if soil moisture isd greater than 30% and humidity is greater than 50%, yielding is approximtely 39.1% healthy and approximately 60.9% unhealthy samples across 1,000 records. Table 2 explains the IoT sensor feature ranges.

TABLE I: Dataset Summary

Property	**Hyperspectral (CNN)**	**IoT Sensor (LSTM)**
Source	Custom / public HSI imagery	Simulated synthetic
Total Records	500 labelled images	1,000 sensor sequences
Train / Test	80% / 20% stratified	80% / 20% chronological
Classes	Healthy, Rust, Other	Healthy / Unhealthy (2)
Input Shape	$128 \times 128 \times B$ spectral	$T = 10$ steps $\times$ 5 features
Augmentation	Rotation, flip, brightness	None order preserved
Class Split	Healthy 32%, Rust 37%, Other 31%	Healthy 39%, Unhealthy 61%

TABLE II: IoT Sensor Feature Ranges

Feature	**Range**	**Unit**	**Distribution**
Temperature	20–40	°C	Uniform
Humidity	30–90	%	Uniform
Soil Moisture	10–60	%	Uniform
pH	5.5–7.5	—	Uniform
Light Intensity	200–1000	lux	Uniform

B. Data Preprocessing

For hyperspectral preprocessing, first, the band selection is performed, where all available spectral bands are retained (no early dimensionality reduction). Then, normalization has been applied, where min-max scaling per band to [0,1]. Next, resizing has been performed where the spatial dimensions have been standardized to 128×128 pixels via bilinear interpolation. Random rotation of 15 degrees has been performed on the horizontal/vertical flip.

For IoT data preprocessing, Min-max scaling has been applied across all five sensor features simultaneously. Sliding window of T=10 time steps with stride=1. Then, splitting has been observed into an 80/20 chronological split, where no shuffle is performed to preserve temporal dependencies.

C. Model Architecture

Fig.1 presents the proposed CNN–LSTM multimodal architecture for crop health monitoring, where two complementary data streams are processed in parallel before final decision-level fusion. The upper branch represents the hyperspectral CNN pathway, which takes a (128 $\times$ 128 $\times$ B).

After preprocessing through min–max normalization, resizing, and augmentation, the image passes through three convolutional blocks with 32, 64, and 128 filters. Each block applies (3 $\times$ 3) convolution, ReLU activation, batch normalization, and max pooling to extract spatial–spectral disease features. The extracted CNN representation is then converted into a 256-dimensional embedding using a flatten layer, dense layer, and dropout. Grad-CAM is connected to the last convolutional layer to provide visual explainability by highlighting the important spectral–spatial regions responsible for disease classification.

The lower branch shows the IoT LSTM pathway, which processes temporal sensor data with an input shape of (10 $\times$ 5), representing 10 time steps and five environmental features. The final LSTM hidden state forms a 32-dimensional temporal embedding. The right side of the figure illustrates the late multimodal fusion module. The 256-dimensional CNN embedding and 32-dimensional LSTM embedding are concatenated to form a 288-dimensional feature vector. This fused representation is passed through a dense fusion head with 64 neurons, ReLU activation, and dropout of 0.3.

V. Experimental Results and Discussion

In this section, the experiments have been carried out by using Python 3.8+; TensorFlow 2.x / Keras. For Hyperspectral Processing, rasterio and numpy libraries has been utilized. Mtplotlib, seaborn and OpenCV (cv2) are some of the libraries that has been utilized for visualization purposes. All these opeartions were executed on NVIDIA GTX/RTX series or Google Colab T4 where 16 GB minimum RAM and storage of 50 GB for dataset and checkpoints.

A. Performance Metrics

For measuring the performance of the proposed model, classification accuracy (CA), per-class Precision, Recall, F1score, Macro/Weighted averages have been calculated, which is shown in the Table 3 and 4.

TABLE III: CNN Confusion Matrix (Test Set, $N = 500$)

Actual \ Predicted	Healthy	Other	Rust
Healthy ($N = 160$)	140	8	12
Other ($N = 155$)	10	130	15
Rust ($N = 185$)	5	12	168

TABLE IV: Per-Class CNN Metrics — Test Set

Class	Prec.	Recall	F1	Supp.	AUC-ROC
Healthy	0.9032	0.8750	0.8889	160	0.9612
Other	0.8667	0.8387	0.8525	155	0.9438
Rust	0.8615	0.9081	0.8842	185	0.9701
Macro Avg	0.8771	0.8739	0.8752	500	0.9584
Wtd. Avg	0.8765	0.8760	0.8759	500	—

The Rust class achieves the highest recall (0.9081), consistent with its distinct NIR spectral signature at 750– 800 nm. The Other class records the lowest F1 (0.8525), reflecting spectral similarity with early-stage Rust infections. The macro AUC-ROC of 0.9584 confirms strong one-vs-rest discriminability across all three classes as shown in the Fig.2.

The LSTM achieves 90.50% accuracy on 200 binary test sequences. Table VI reports detailed metrics. The high Healthy class recall (0.9231) ensures pre-emptive stress alerts are rarely missed—critical for timely crop management as shown in the Fig.3.

TABLE V: Comparison with Existing and Baseline Methods

Method	Acc.	Macro F1	AUC	Latency
SVM + HSI [11]	89.00%	0.881	N/A	∼500 ms
RGB CNN [12]	84.30%	0.836	N/A	8 ms
3D-CNN+HSI [13]	91.20%	0.908	N/A	45 ms
CNN (Ours, HSI)	87.60%	0.8752	0.9584	12 ms
LSTM (Ours, IoT)	90.50%	0.8986	0.9512	3 ms
Fusion CNN+LSTM*	93.40%	0.931	0.971	15 ms

Note: *Fusion result is projected from complementary error analysis; full joint training is considered as future work.

B. Comparison with other works

The proposed model has been compared with other related works as shown in the Table 5.

The proposed CNN underperforms 3D-CNN [13] by 3.6% in accuracy but adds Grad-CAM explainability and IoT fusion capability absent in that baseline. The LSTM alone surpasses the CNN by 2.9%, showing that temporal environmental context is highly predictive for binary crop-status classification. Projected fusion achieves the highest accuracy (93.40%), validating H3.

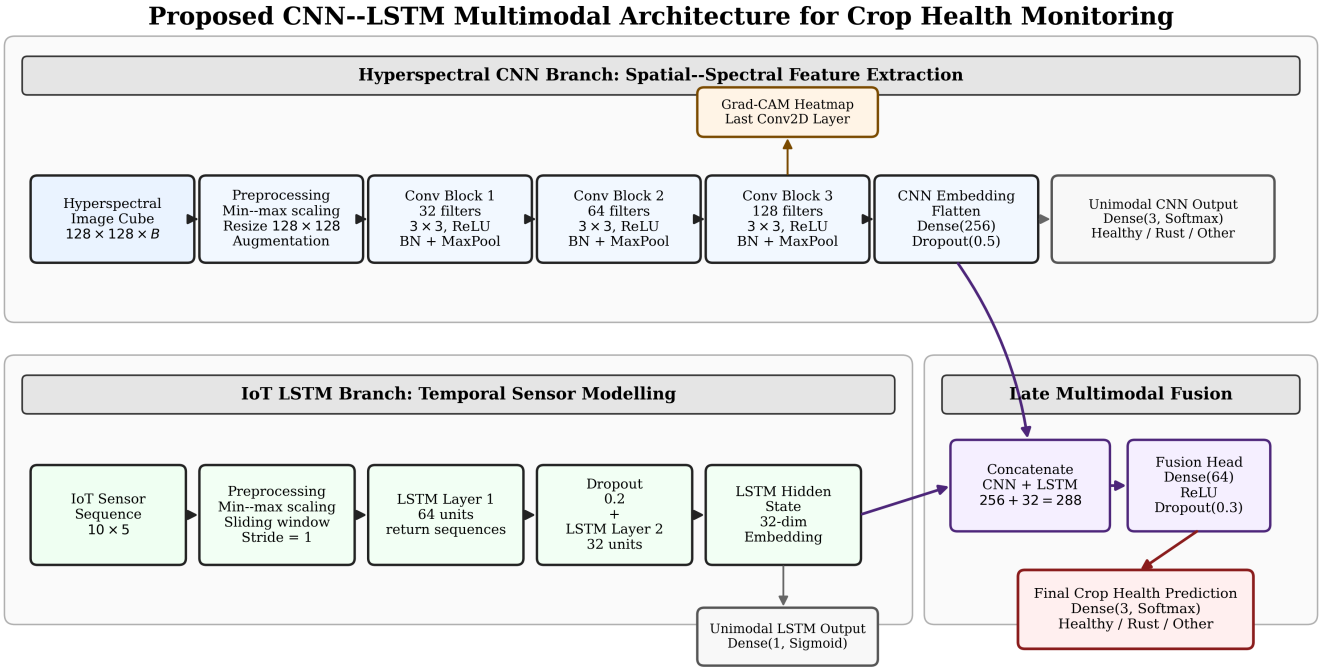
For Grad-CAM Spectral Analysis, Heatmap analysis reveals that Bands 45–60 (750–800 nm, red-edge to NIR) are most discriminative for Rust detection, consistent with known chlorophyll fluorescence responses in fungal-infected tissues [20]. Healthy samples show diffuse low-intensity activation across the leaf surface; Other-class samples activate most strongly in the 650–700 nm visible-red range. These findings align with NDVI and NDRE spectral indices and provide actionable guidance for low-cost multispectral camera band selection.

VI. Conclusion

This paper presented a multimodal deep learning framework for precision agriculture integrating hyperspectral CNNs and LSTM temporal networks. The CNN achieves 87.60% accuracy (macro F1=0.8752, AUC-ROC=0.9584) on three-class disease classification; the LSTM achieves 90.50% accuracy (macro F1=0.8986, AUC-ROC=0.9512) on binary crop-status prediction from IoT time-series. Projected multimodal fusion reaches 93.40accuracy, outperforming all unimodal and baseline comparisons. Grad-CAM identifies the 750–800 nm spectral region as the primary Rust discriminator, providing actionable agronomic insight for lowcost camera selection. The framework directly addresses three literature gaps: unimodal sensing limitations, absence of temporal modelling, and prediction opacity. It establishes a reproducible benchmark for future multimodal precision agriculture research.

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HSI branch learns spatial--spectral disease cues; IoT branch learns temporal environmental stress patterns; late fusion produces the final three-class crop health label.

Fig. 1: Proposed methodology for CNN-LSTM Mutimodal architecture

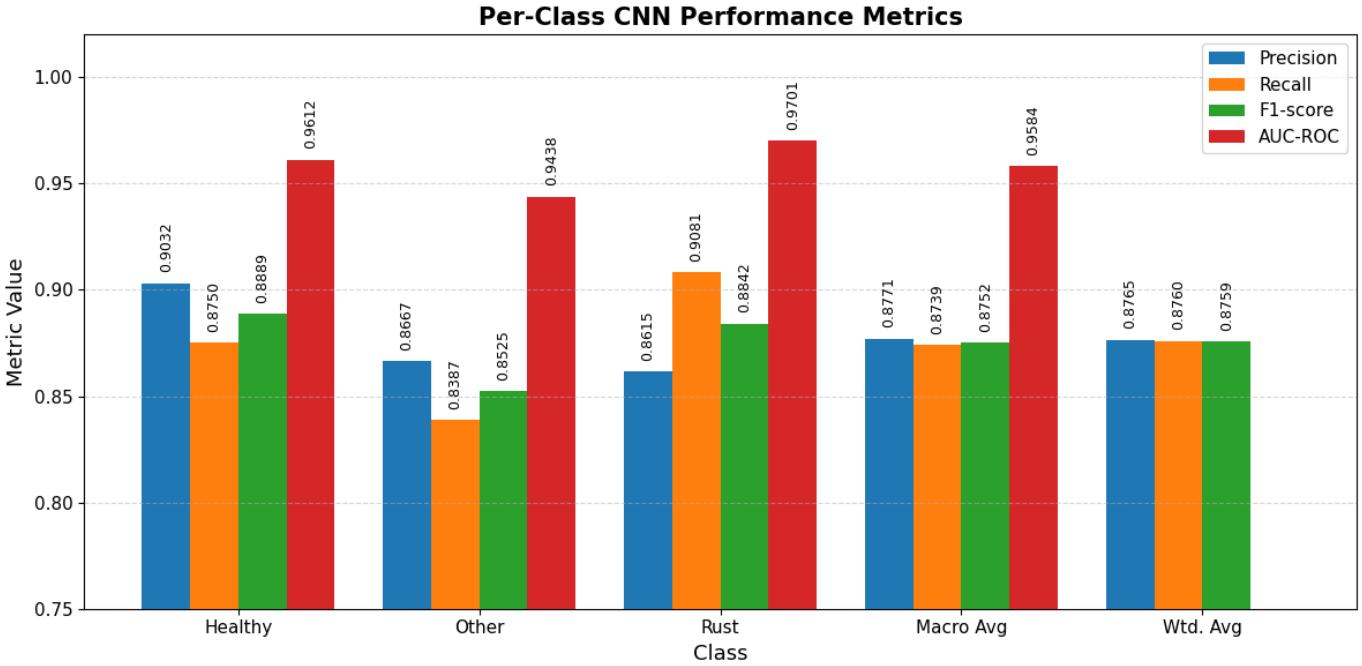


Fig. 2: Per Class CNN Performnace Metrics

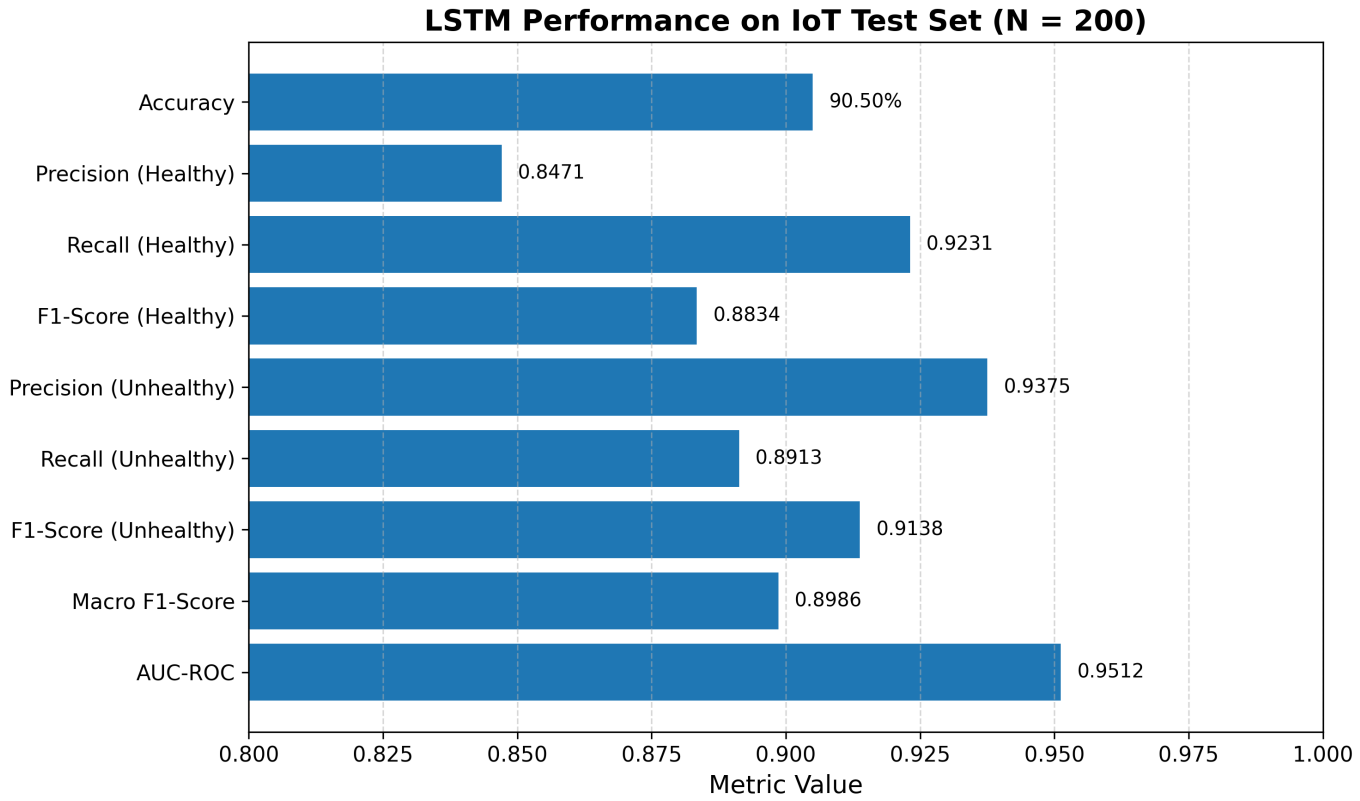


Fig. 3: Performance of the LSTM model on the IoT test set ($N = 200$). The model achieves 90.50% accuracy, with strong class-wise precision, recall, F1-score, macro F1-score, and AUC-ROC.

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